

FLEX And RIGID-FLEX PCBs



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Reliability is important for electronic devices. Hence, a substantial number of electronics' manufacturers focus on the reputation of their brand. For example, connectors often come with deficient performance and can affect overall performance of cable assemblies.

One solution for avoiding quality and reliability issues arising due to connectors and cables is to use flexible solutions, especially for printed circuit boards (PCBs). Rigid-flex PCBs are ideal for connecting two or more PCBs.

Benefits of using rigid-flex PCBs

Various benefits of rigid-flex PCBs are given below.

Repeatable installation and wire routing. Flex PCBs can be compared to discrete wiring, or ribbon cables. These eliminate wire routing errors, and reduce text time, reworks, rejects and service calls.

Long duty cycles. Depending on the buildup, flex circuits can be thin, robust and enough for a few million bending flexion cycles without causing any damages.

High vibration and robust connections. Under vibration and/or high acceleration, low mass and ductility of flex PCBs reduce soldering joints failures.

Space and weight reduction. Flex circuits can replace several wires, cables, hand boards

TABLE I PI FLEX FOILS VERSUS RIGID FR4			
Flex vs Rigid	Cu Peel strength (N/cm)	Shrinkage after etching (0%)	Water absorption (0%)
Flexible material PI	>7	0~0.2	2.9
Rigid base material FR4	14	0~0.2	0.5

and connectors in a minimum space area. Flex PCBs save up to 60 per cent weight and space, compared to rigid PCBs.

Fast assembly. Flex PCBs eliminate the need for colour codes and wrap bundles of wires, reducing the chances of assembly failures. These enable lower mounting times for safety in production, especially for large volumes.

Flexible. A combination of design freedom, space, weight and components saving can reduce packaging requirements significantly when using flex circuits.

Cost effective. Flex PCBs require a smaller area and fewer parts for the final assembled product.

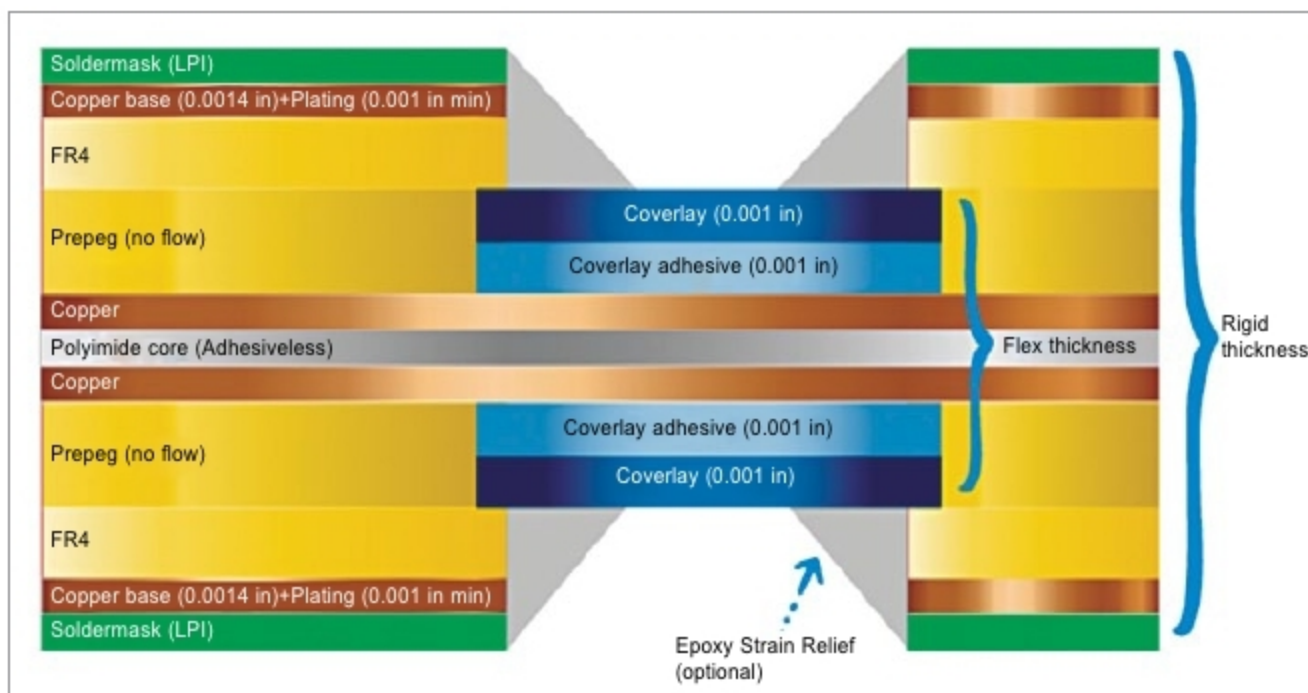
Reliability and durability. Flex circuits can be built with materials that are suitable to use in harsh operating environments. Reliability can be increased by eliminating interface connections (solder joints, connectors, etc). Flex boards cover the conductors with polyimide. This dielectric protection is good against ambient conditions and environments.

Different flex PCB structures

There are two structures that can be followed for flex PCB development: symmetrical and unsymmetrical.

Symmetrical structure. Buildup is symmetrical in z axis. Several flex parts can be laminated with or without air gap. This is the most common structure used for rigid-flex boards.

Rigid-flex circuits combine the best of both—rigid boards and flexible circuits—integrated together into one circuit



Standard buildup for rigid-flex PCBs